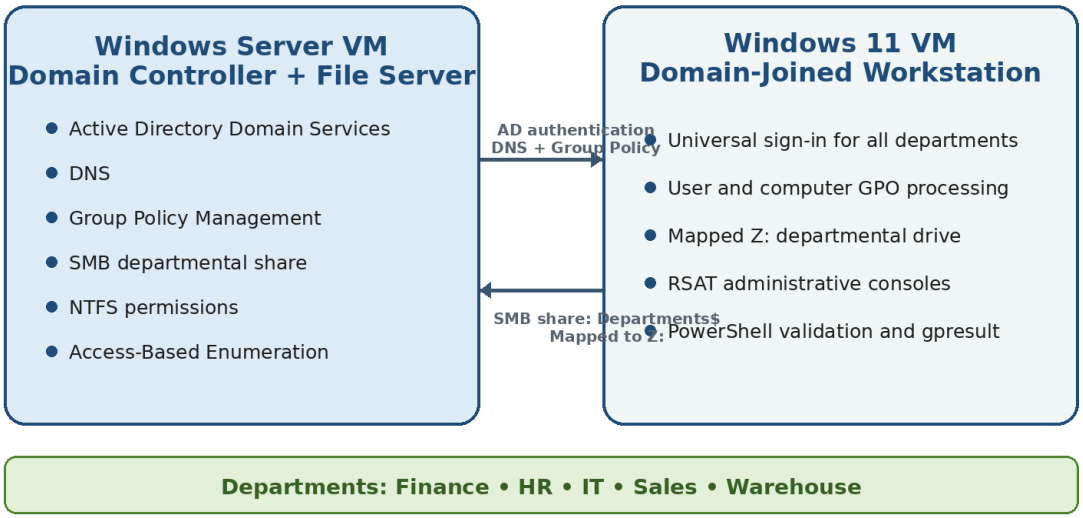
WINDOWS SERVER ACTIVE DIRECTORY LAB

Technical Project Write-Up

Two-VM enterprise-style environment for identity, policy, access control, and departmental file services

Two-VM Active Directory Lab Architecture

One Windows Server VM provides core domain services; one Windows 11 VM acts as the universal domain workstation.



Prepared by Nicholas Sugrim
Self-Directed Lab • July 2026

Executive Summary

This project is a two-virtual-machine Windows domain designed to demonstrate the core identity, access, policy, and troubleshooting tasks commonly found in entry-level systems administration and IT support environments. One Windows Server virtual machine provides Active Directory Domain Services, DNS, Group Policy administration, and departmental SMB file storage. A Windows 11 workstation is joined to the domain and acts as a universal endpoint where users from Finance, HR, IT, Sales, and Warehouse can sign in and receive the correct policies and resources.

The environment uses organizational units to separate users, computers, role groups, and resource groups. Access is assigned through an AGDLP-style model: users are placed in Global role groups, the role groups are nested into Domain Local resource groups, and the Domain Local groups receive NTFS permissions. This design allows Sales users to read Warehouse inventory reports without receiving permission to modify the source files.

Group Policy is divided by purpose so that domain account policies, workstation security, workstation auditing, mapped drives, and department-specific user settings can be tested independently. A delegated IT support role was also configured to perform routine account tasks and review policy configuration without receiving local administrator or Domain Admin rights.

Category	Implementation
Project type	Self-directed Windows Server and Active Directory lab
Virtual machines	1 Windows Server VM and 1 Windows 11 VM
Domain	ADLab.Domain
Departments	Finance, HR, IT, Sales, and Warehouse
Core services	AD DS, DNS, Group Policy, SMB file sharing, NTFS permissions
Administration tools	ADUC, GPMC, RSAT, Event Viewer, Computer Management, PowerShell

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1. Project Goals and Design Constraints

The primary objective was to build a portfolio-ready Active Directory environment that demonstrates practical administration rather than a simple domain-join exercise. The lab was intentionally limited to two virtual machines: one Windows Server domain controller that also hosts departmental file services, and one Windows 11 workstation that supports sign-in for users from every department.

The design goals were:

- Create a clean OU structure for users, computers, security groups, disabled accounts, and departmental identities.
- Use role-based access control instead of assigning permissions directly to individual users.
- Apply both computer and user Group Policies to a shared workstation.
- Map a common Z: drive while allowing Access-Based Enumeration and NTFS permissions to control what each user sees.
- Demonstrate cross-department access, including Sales read-only access to Warehouse inventory reports.
- Delegate routine account administration without granting broad administrative privileges.
- Produce repeatable validation evidence through graphical consoles and PowerShell reporting.

Production note: combining the domain controller and file server is acceptable for this constrained lab, but a production design would normally separate those roles and deploy at least two domain controllers for redundancy.

2. Lab Architecture

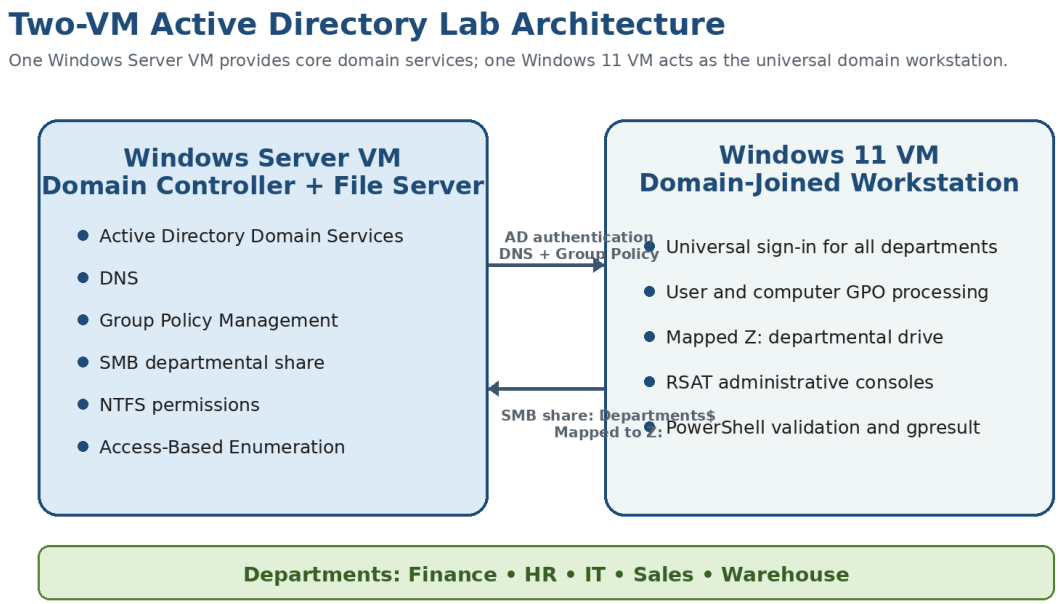


Figure 1. Two-VM Active Directory lab architecture.

Component	Responsibilities
Windows Server VM	Domain controller, AD DS, DNS, Group Policy Management, SMB departmental share, NTFS permissions, and Access-Based Enumeration.
Windows 11 VM	Domain-joined universal workstation, user and computer policy processing, mapped-drive testing, RSAT consoles, Event Viewer, and PowerShell validation.

Component	Responsibilities
Department share	The server-local Z:\ volume is shared as Departments$ and mapped to Z: on the workstation. The share root contains Finance, HR, IT, Sales, and Warehouse.

3. Active Directory and OU Design

The domain uses a dedicated Lab OU to keep project objects separate from default Active Directory containers. Computers, groups, and users are separated so that policy linking, delegation, and administration remain easy to understand and troubleshoot. The workstation computer object is placed in the workstation OU so that baseline and auditing policies apply to the machine, while departmental users remain in their individual user OUs and receive identity-specific settings on the same endpoint.

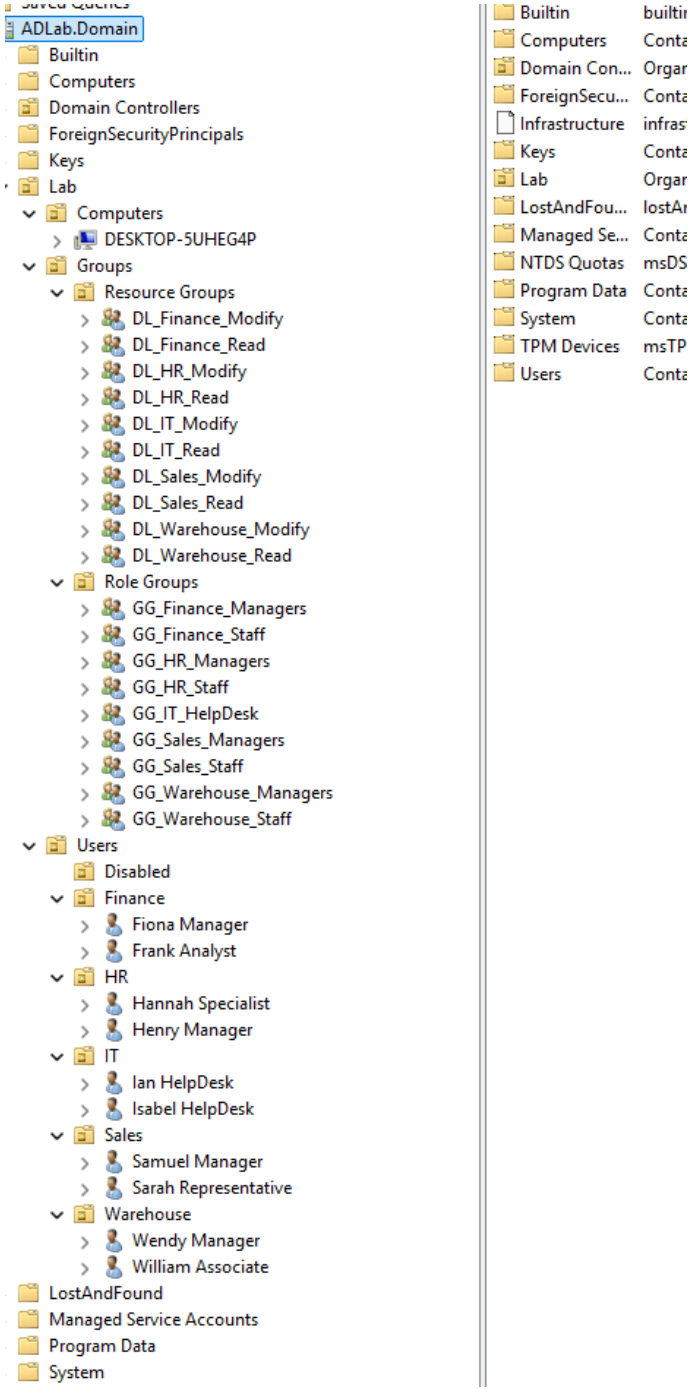


Figure 2. Active Directory OU hierarchy used by the lab.

OU / Container	Purpose
Lab\Computers\Server	Contains the Windows Server computer object.
Lab\Computers\Workstation	Contains the domain-joined Windows workstation and receives computer GPOs.
Lab\Groups\Role Groups	Stores Global security groups that describe department or job role.
Lab\Groups\Resource Groups	Stores Domain Local security groups associated with resource permissions.
Lab\Users\Departments	Parent OU for Finance, HR, IT, Sales, and Warehouse user accounts.
Lab\Users\Disabled	Controlled location for inactive or terminated accounts.

This structure supports clear separation between user policy and computer policy. User GPOs are linked to departmental OUs, while workstation security and auditing GPOs are linked to the workstation computer OU. Loopback processing is not required because the correct user settings follow the user account and the correct computer settings follow the workstation object.

4. Identity, Security Groups, and RBAC

The lab uses an AGDLP-style access model. Accounts are placed in Global groups representing their business role. Those role groups are nested into Domain Local groups representing a permission on a specific folder. NTFS permissions are assigned to the Domain Local groups, not directly to users.

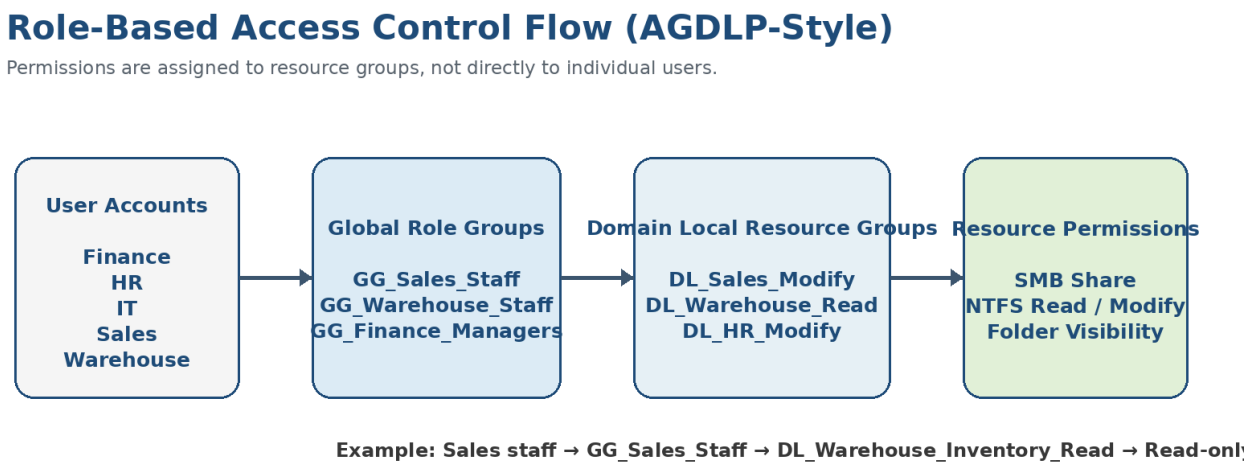


Figure 3. Role-based permission flow from users to resources.

```
Fiona Manager (fiona.manager)
  DL_Finance_Modify
  GG_Finance_Managers

Frank Analyst (frank.analyst)
  DL_Finance_Read
  GG_Finance_Staff

Hannah Specialist (hannah.specialist)
  DL_HR_Read
  GG_HR_Staff

Henry Manager (henry.manager)
  DL_HR_Modify
  GG_HR_Managers

Ian HelpDesk (ian.helpdesk)
  DL_IT_Modify
  GG_IT_HelpDesk

Isabel HelpDesk (isabel.helpdesk)
  DL_IT_Modify
  GG_IT_HelpDesk

Samuel Manager (samuel.manager)
  DL_Sales_Modify
  GG_Sales_Managers

Sarah Representative (sarah.representative)
  DL_Sales_Read
  GG_Sales_Staff

Wendy Manager (wendy.manager)
  DL_Warehouse_Modify
  GG_Warehouse_Managers

William Associate (william.associate)
  DL_Warehouse_Read
  GG_Warehouse_Staff
PS C:\Users\ADLabAdmin>
```

Figure 4. Example user membership in a departmental Global security group.

Group type	Example	Purpose
Global role group	GG_Sales_Staff	Collects users who perform the same departmental role.
Global manager group	GG_Finance_Managers	Identifies managers for targeted shortcuts and management-folder access.
Domain Local Modify group	DL_Sales_Modify	Receives Modify permission on the Sales department resource.
Domain Local Read group	DL_Warehouse_Inventory_Read	Receives Read and Execute permission on Warehouse inventory reports.

Cross-department example: Sales users belong to GG_Sales_Staff. That Global group is nested into DL_Warehouse_Inventory_Read, which receives Read and Execute permission on the Warehouse inventory-report folder. Warehouse staff remain members of a separate Modify resource group, allowing them to create, edit, and delete reports while Sales can only view or copy them.

5. Departmental File Services

The server-local data drive is organized directly beneath Z:\ with one top-level folder for each department. The root of the volume is shared as Departments$, so the workstation mapping presents the same directory structure as Z:\Finance, Z:\HR, Z:\IT, Z:\Sales, and Z:\Warehouse.

Server path	Workstation path	Intended access
Z:\Finance	Z:\Finance	Finance staff; manager subfolder restricted to Finance managers.
Z:\HR	Z:\HR	HR staff; employee records and management resources protected from other departments.
Z:\IT	Z:\IT	IT documentation, scripts, and software repository.
Z:\Sales	Z:\Sales	Sales staff with Modify rights; management resources restricted to Sales managers.
Z:\Warehouse	Z:\Warehouse	Warehouse staff with Modify rights; inventory reports readable by Sales.

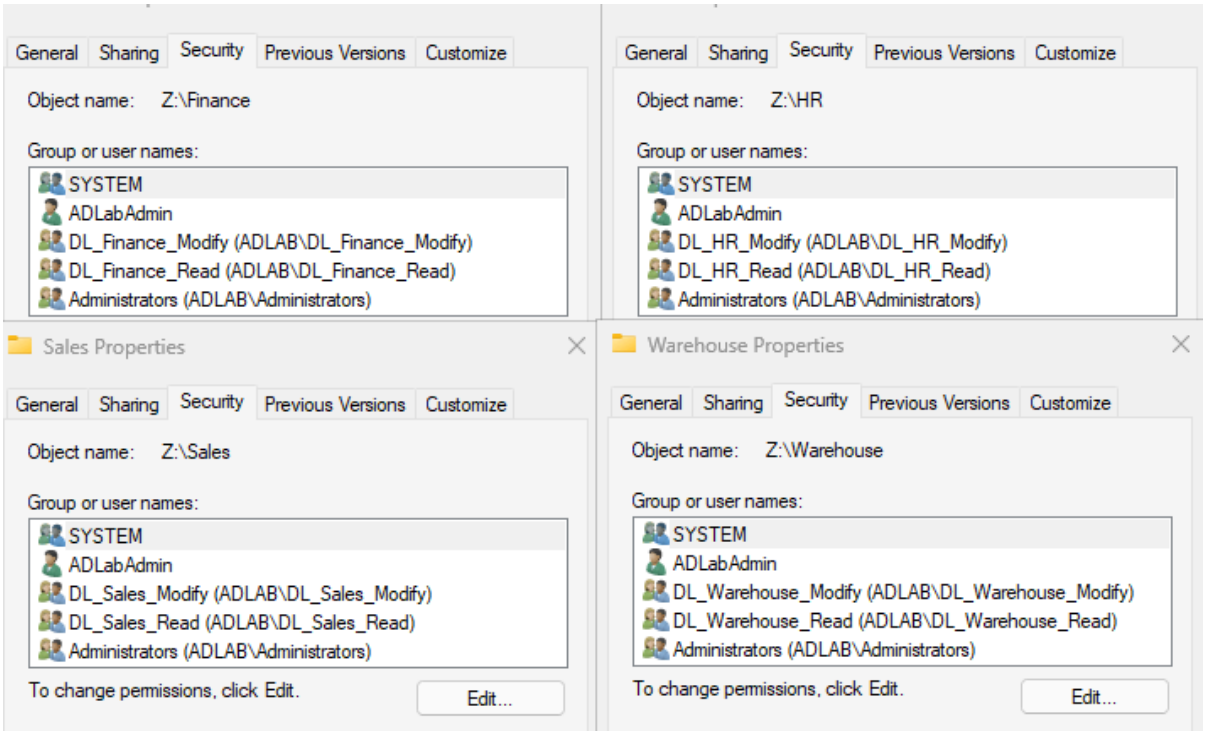


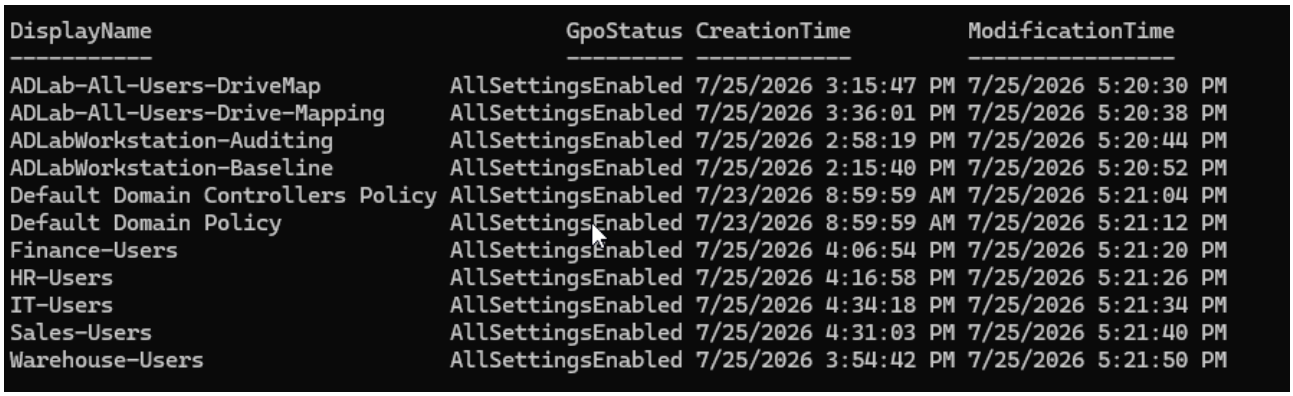
Figure 5. NTFS access assigned to Domain Local resource groups.

Share permissions provide the ability to connect to the SMB share, while NTFS permissions provide the detailed authorization. Access-Based Enumeration is enabled so users do not see folders for which they lack access. Broad principals such as Everyone or Domain Users are not granted Modify access to departmental content.

The lab includes realistic fictional files for each department, including inventory reports, finance workbooks, HR forms, sales quotes, IT documentation, and validation scripts. This makes permission testing visible and repeatable rather than relying on empty folders.

6. Group Policy Architecture

Group Policy is separated by function. Domain account policies remain at the domain level, computer security and auditing policies are linked to the workstation OU, the common mapped-drive policy is linked to the parent user hierarchy, and each department receives a dedicated user policy.



```
DisplayName                          GpoStatus          CreationTime         ModificationTime
-----------                          ---------          ------------         ----------------
ADLab-All-Users-DriveMap             AllSettingsEnabled 7/25/2026 3:15:47 PM 7/25/2026 5:20:30 PM
ADLab-All-Users-Drive-Mapping        AllSettingsEnabled 7/25/2026 3:36:01 PM 7/25/2026 5:20:38 PM
ADLabWorkstation-Auditing            AllSettingsEnabled 7/25/2026 2:58:19 PM 7/25/2026 5:20:44 PM
ADLabWorkstation-Baseline            AllSettingsEnabled 7/25/2026 2:15:40 PM 7/25/2026 5:20:52 PM
Default Domain Controllers Policy    AllSettingsEnabled 7/23/2026 8:59:59 AM 7/25/2026 5:21:04 PM
Default Domain Policy                AllSettingsEnabled 7/23/2026 8:59:59 AM 7/25/2026 5:21:12 PM
Finance-Users                        AllSettingsEnabled 7/25/2026 4:06:54 PM 7/25/2026 5:21:20 PM
HR-Users                             AllSettingsEnabled 7/25/2026 4:16:58 PM 7/25/2026 5:21:26 PM
IT-Users                             AllSettingsEnabled 7/25/2026 4:34:18 PM 7/25/2026 5:21:34 PM
Sales-Users                          AllSettingsEnabled 7/25/2026 4:31:03 PM 7/25/2026 5:21:40 PM
Warehouse-Users                      AllSettingsEnabled 7/25/2026 3:54:42 PM 7/25/2026 5:21:50 PM
```

Figure 6. Group Policy inventory for the domain, workstation, drive mapping, and departments.

6.1 Common Department Drive Mapping

Setting	Configuration
Policy path	User Configuration > Preferences > Windows Settings > Drive Maps
Action	Update
Location	\\<domain-controller>\Departments$
Drive letter	Z:
Label	Department Files
Options	Reconnect enabled; remove the mapping when the item no longer applies.
Scope	Parent Lab user hierarchy so all departmental users inherit the mapping.

Every user receives the same root mapping. Department visibility and access level are controlled by Access-Based Enumeration and NTFS permissions rather than by creating a different drive letter for each department. One duplicate drive-map GPO was retained temporarily during troubleshooting; the intended final configuration uses one authoritative Z: mapping to avoid duplicate preference processing.

6.2 Workstation Security Baseline

Policy setting	Configured value
Interactive logon: Do not display last signed-in user	Enabled
Interactive logon: Machine inactivity limit	900 seconds
Interactive logon message title	ADLab Authorized Access
Interactive logon message text	Authorized-use and activity-monitoring notice
Accounts: Guest account status	Disabled
UAC: Run administrators in Admin Approval Mode	Enabled
UAC: Switch to secure desktop when prompting	Enabled
Microsoft Defender Antivirus	Real-time protection, behavior monitoring, downloaded-file scanning, and program-activity monitoring enabled
Windows Defender Firewall	Domain, Private, and Public profiles enabled; inbound blocked and outbound allowed by default
Firewall logging	Dropped-packet logging enabled; 16,384 KB maximum log size
Automatic Updates	Automatic download and scheduled installation configured

6.3 Workstation Auditing

Audit area	Configured events
Advanced audit policy override	Enabled so subcategory settings override legacy audit categories.
Logon and Logoff	Logon success/failure, logoff success, account lockout failure, and special logon success.
Account Management	User-account, security-group, and computer-account management success/failure.
Detailed Tracking	Process creation success with command-line data included in process events.
Object Access	File-share success/failure, detailed file-share failure, and removable-storage success/failure.
PowerShell logging	Script Block Logging enabled; Module Logging enabled for all modules.

6.4 Domain Password and Lockout Policy

Setting	Value
Password history	10 passwords remembered
Maximum password age	90 days
Minimum password age	1 day
Minimum password length	12 characters
Password complexity	Enabled
Reversible encryption	Disabled
Account lockout threshold	5 invalid attempts
Lockout duration	15 minutes
Reset lockout counter	15 minutes

The Default Domain Controllers Policy remains linked to the Domain Controllers OU for controller-specific settings and is kept separate from workstation and departmental configuration.

7. Department-Specific User Policies

Each department receives a user GPO linked to its OU. Group Policy Preferences create shortcuts to relevant resources, while user-side Personalization settings enforce department-appropriate inactivity controls. Manager shortcuts use item-level targeting, but NTFS permissions remain the authoritative security boundary.

GPO	Shortcuts and resources	Security / personalization
Finance-Users	Finance Files (Z:\Finance); Financial Reports; manager shortcut targeted to GG_Finance_Managers.	Screen saver enabled; password required on resume; 300-second timeout; users cannot change screen saver.
HR-Users	HR Files (Z:\HR); Employee Records; manager shortcut targeted to GG_HR_Managers.	Screen saver enabled; password required on resume; 300-second timeout; users cannot change screen saver.
Sales-Users	Sales Files; Customer Quotes; Warehouse Inventory - Read Only; manager shortcut targeted to GG_Sales_Managers.	Screen saver enabled; password required on resume; 900-second timeout.
Warehouse-Users	Warehouse Files; Inventory Reports; manager shortcut targeted to GG_Warehouse_Managers.	Screen saver enabled; password required on resume; 900-second timeout.
IT-Users	IT Documentation, IT Scripts, IT Software, ADUC, GPMC, Event Viewer, and Computer Management.	Provides centralized tools and resources without granting additional permissions by itself.

8. Delegated IT Support Administration

A least-privilege IT support role was created through a Global security group. The account uses RSAT from the domain workstation and receives delegated rights over normal departmental users, while privileged accounts remain outside the delegated OU scope.



Figure 7. Delegation of Control applied to the IT support security group.

Allowed task	Implementation
View normal user accounts	Read all user information delegated on the departmental user hierarchy.
Reset passwords	Reset user passwords and force password change at next logon delegated through ADUC.
Unlock accounts	User-account unlock permission delegated for normal departmental identities.
Edit approved user fields	Description, office, phone, mobile, address, title, department, and company.
Review Group Policy	Read-only permissions on GPOs; optional Group Policy Results access for troubleshooting.

The delegated role is intentionally not a member of Domain Admins, Account Operators, Group Policy Creator Owners, or the workstation local Administrators group. It cannot create or delete users, change group membership, edit or link GPOs, reset privileged administrator passwords, or automatically access sensitive HR and Finance files.

9. RSAT and Administrative Tooling

Remote Server Administration Tools were installed on the Windows workstation so routine administration and troubleshooting could be performed without interactive administration of the domain controller. Installing RSAT provides the consoles but does not grant permissions; Active Directory delegation and GPO permissions determine what each account can do.

RSAT feature / shortcut	Purpose
Active Directory Domain Services and LDS Tools	ADUC, Active Directory Administrative Center, and directory-management components.
Group Policy Management Tools	View GPO settings, links, and Group Policy Results.
DNS Server Tools	Inspect DNS zones and records during name-resolution troubleshooting.
Server Manager	View server roles and management status from the workstation.
Event Viewer	Review System, Application, Security, and GroupPolicy Operational logs.
Computer Management	Review local services, devices, shared folders, and workstation state.
IT Documentation / Scripts / Software	Shortcuts to centralized IT resources on Z:\IT.

10. Validation and Testing

The configuration was tested from the domain workstation using separate user identities. Group membership changes were followed by a complete sign-out and sign-in so the user security token contained the updated memberships.

- Verified that the workstation receives baseline and auditing GPOs after its computer object is placed in the workstation OU.
- Confirmed that all departmental users receive the Z: mapping and only see folders authorized through NTFS permissions and Access-Based Enumeration.
- Confirmed that Finance, HR, IT, Sales, and Warehouse access is granted through role-group and resource-group nesting rather than direct user permissions.
- Verified that Sales can read Warehouse inventory reports but cannot create, edit, or delete files in the protected folder.
- Verified that manager-only shortcuts appear only for the corresponding manager Global groups.
- Confirmed that department-specific screen-lock settings and shortcuts change according to the user who signs into the shared workstation.
- Confirmed that the delegated IT support account can reset and unlock normal users but cannot edit GPOs or administer privileged identities.
- Reviewed applied user and computer policy through Group Policy Results and HTML reports generated by gpresult.

11. PowerShell Reporting and Evidence

Although the environment was configured primarily through graphical tools, PowerShell was used to inventory and validate the finished design. These commands produce reproducible evidence that can be included in a portfolio or used during troubleshooting.

```
# Active Directory structure and identities
Get-ADOrganizationalUnit -Filter *
Get-ADUser -Filter * -Properties Department,Title,Enabled
Get-ADGroup -Filter *
Get-ADGroupMember 'DL_Warehouse_Inventory_Read'
Get-ADPrincipalGroupMembership 'it.helpdesk'

# Group Policy and workstation results
Get-GPO -All
gpresult /h C:\Temp\GPO-Results.html /f

# File services and access
Get-SmbShare
Get-SmbShareAccess -Name 'Departments$'
Get-SmbMapping
Get-Acl 'Z:\Warehouse\Inventory Reports'
```

Recommended portfolio outputs include the OU inventory, GPO inventory, key group-nesting reports, mapped-drive status, SMB-share configuration, NTFS access-control entries, the workstation local Administrators membership, and a gpresult HTML report for at least one department user.

12. Results, Skills Demonstrated, and Future Improvements

The completed environment represents a compact departmental Windows domain. One workstation can serve users from five departments while applying different shortcuts, inactivity controls, and resource permissions according to the signed-in identity. Centralized policy and role-based permissions make the configuration easier to audit and maintain than direct per-user assignments.

Skills demonstrated:

- Active Directory Domain Services and DNS
- OU design and computer-object placement
- Global and Domain Local security groups
- AGDLP-style RBAC and least privilege
- SMB shares, Access-Based Enumeration, and NTFS permissions
- User and computer Group Policy processing
- Group Policy Preferences and item-level targeting
- Password and account-lockout policy
- Windows Defender, Firewall, auditing, and PowerShell logging
- RSAT, delegated administration, and Group Policy Results
- PowerShell inventory and validation

Future improvements for a production-style expansion:

- Add a second domain controller for redundancy and DNS resilience.
- Separate the file-server role from the domain controller.
- Deploy Windows LAPS for local administrator password management.
- Add centralized backups and test restoration procedures.
- Introduce DHCP, patch-management, and software-deployment services.
- Forward security and PowerShell logs to a centralized monitoring platform.

Appendix A. GPO Configuration Matrix

GPO	Scope	Primary purpose
Default Domain Policy	Domain root	Password history, age, length, complexity, and account lockout controls.
Default Domain Controllers Policy	Domain Controllers OU	Controller-specific default security assignments.
ADLabWorkstation-Baseline	Workstation computer OU	Interactive logon, inactivity lock, UAC, Defender, Firewall, and Updates.
ADLabWorkstation-Auditing	Workstation computer OU	Authentication, account-management, process, file-share, removable-storage, and PowerShell logging.
ADLab-All-Users-DriveMap	Parent user hierarchy	Maps Departments$ to Z: for all departmental users.
Finance-Users	Finance OU	Finance shortcuts and five-minute protected screen saver.
HR-Users	HR OU	HR shortcuts and five-minute protected screen saver.
Sales-Users	Sales OU	Sales resources, read-only Warehouse inventory shortcut, and 15-minute protected screen saver.
Warehouse-Users	Warehouse OU	Warehouse resources and 15-minute protected screen saver.
IT-Users	IT OU	IT resources and administrative-console shortcuts.

Appendix B. Validation Test Matrix

Test identity	Expected outcome	Result to capture
Warehouse staff	Modify Warehouse resources and inventory reports.	Create, edit, and delete a test file.
Sales staff	Modify Sales resources; read Warehouse inventory reports.	Open report successfully; modification or deletion denied.
Finance staff	Access Finance resources; no access to HR resources.	Visible Finance folder and denied HR path.
HR staff	Access HR resources; no access to Finance or Sales.	Visible HR folder and denied unauthorized paths.
Department manager	Receive the department management shortcut and folder access.	Shortcut visible only after manager group membership.
IT support account	Reset/unlock normal users and read GPOs.	Successful user task; GPO edit unavailable or denied.
Privileged administrator	Perform domain and GPO administration.	Used only for administrative changes, not routine sign-in.

End of Project Write-Up